

AWS re:Invent

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NET213

How to build network observability for better experiences

Richi Kumari

Sr Product Manager
Amazon Web Services

Dennis Marinus

Sr Software Developer
Amazon Web Services



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CloudWatch internet performance monitoring

PROVIDES VISIBILITY INTO HOW INTERNET ISSUES IMPACT APPLICATION PERFORMANCE, REDUCING THE TIME IT TAKES DEVELOPERS TO DIAGNOSE ISSUES FROM DAYS TO MINUTES

CHALLENGE

Customers lack visibility into how the performance of internet service providers (ISPs) impacts the experience of end users connecting to their AWS hosted applications over the internet

Today, developers spend days diagnosing whether the internet or the application is the cause of an application performance issue



SOLUTION

End-to-end visibility into the internet performance of your application



Internet clients

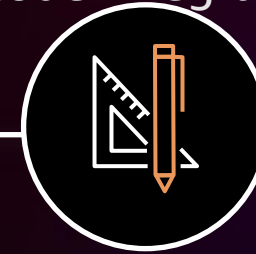


Public internet



Customer applications
(AWS Regions)

Quickly understand the internet-related problems that your end users are experiencing in different geographic locations and networks, without needing any code integration or agents



User personas



Application developer

- Rule out the internet as the cause of an issue within minutes
- Avoid debugging application unnecessarily



DevOps/Network operator

- Monitor networks continuously and understand application impact
- Route users to alternate endpoints

Amazon CloudWatch Internet Monitor story

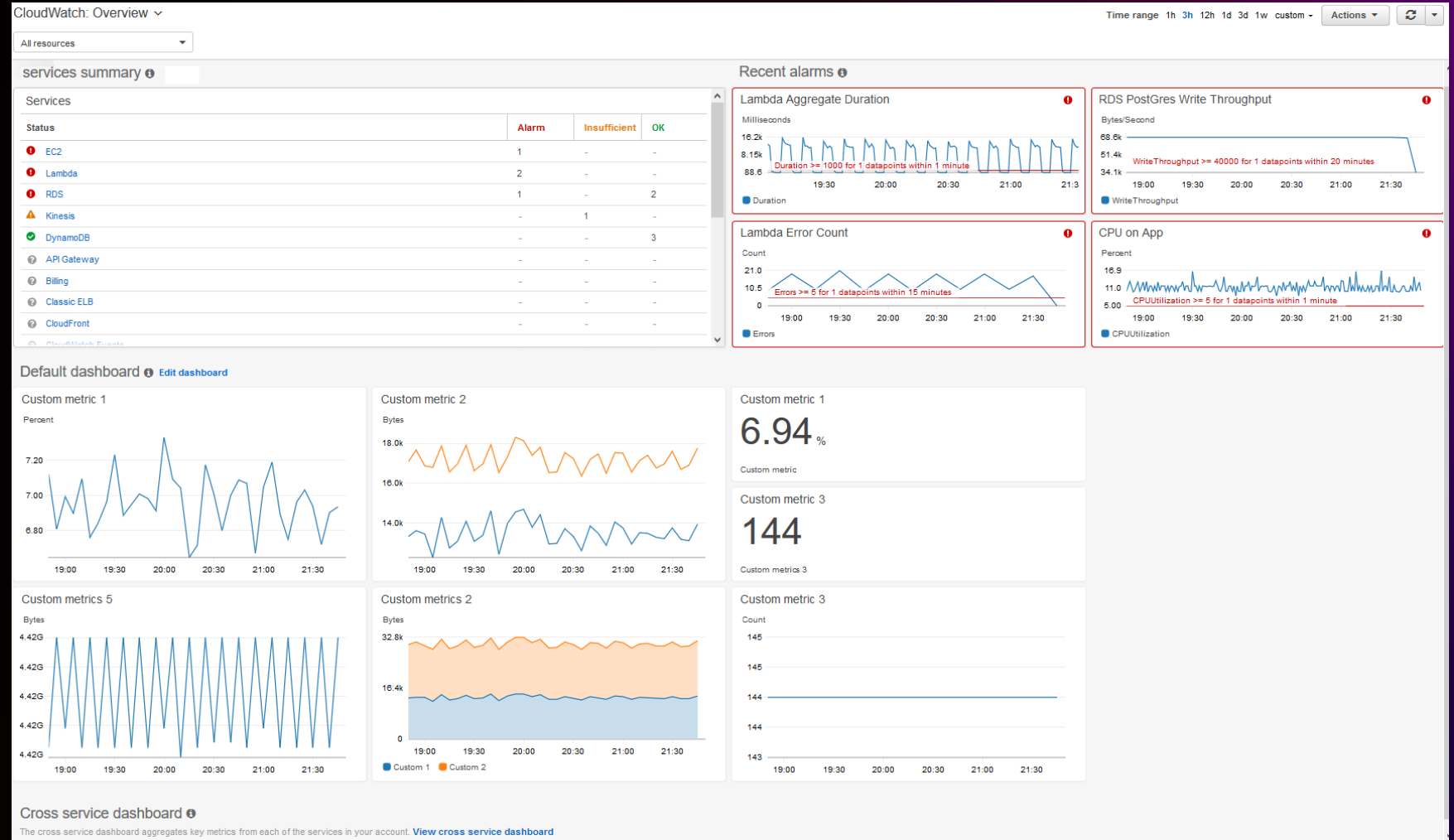


Meet Sam



Sam is an application developer at a popular company

Sam is being a good ops engineer



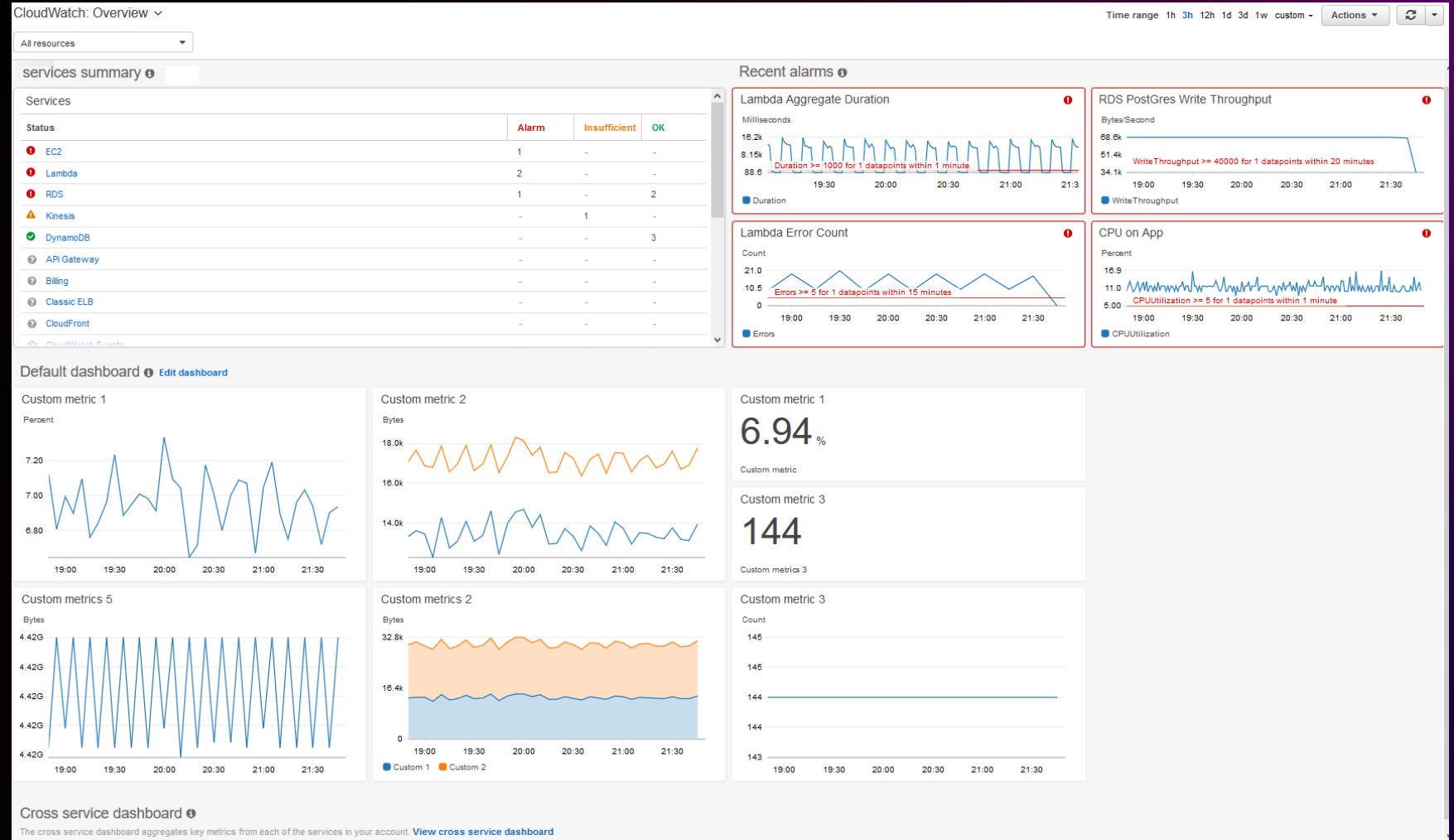
But then this happens



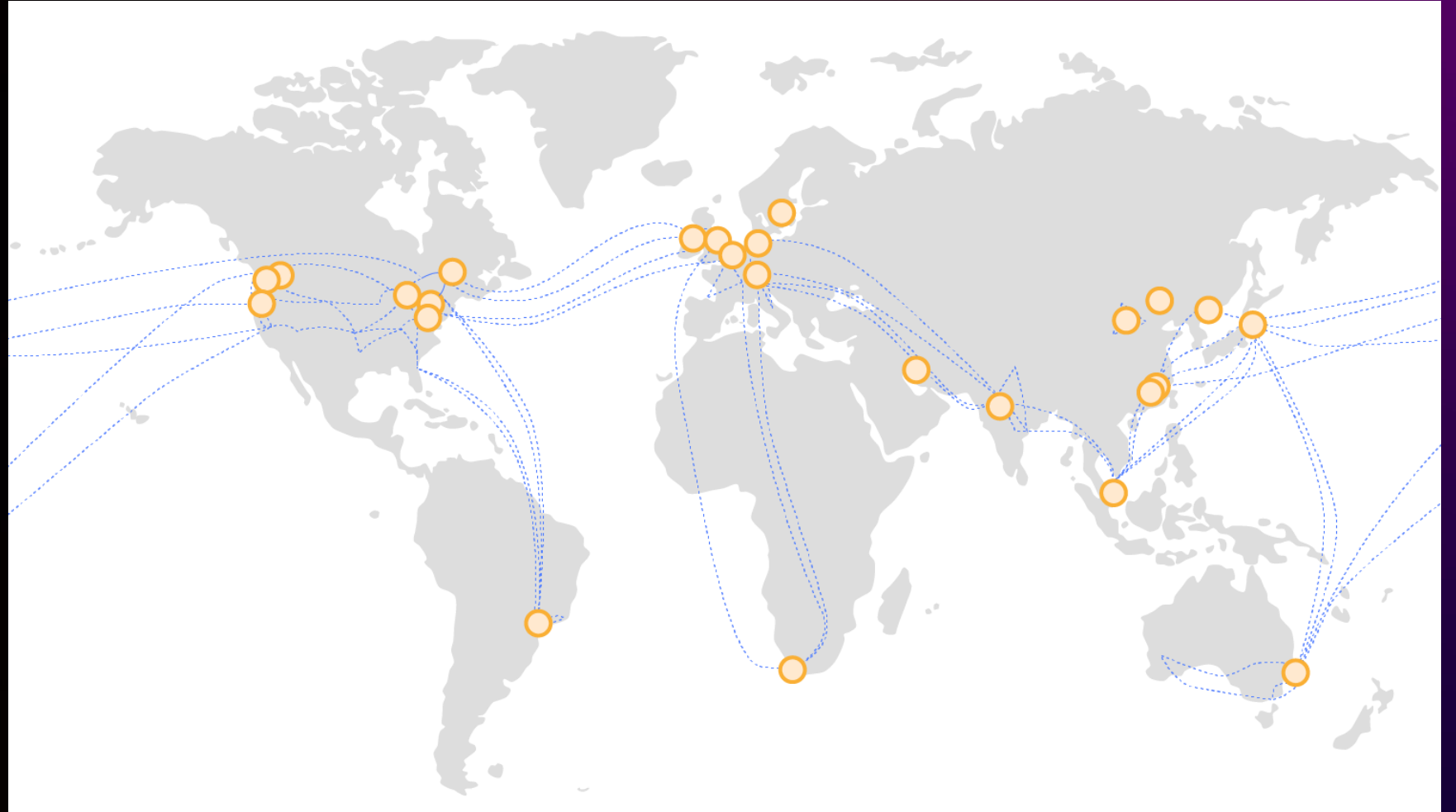
From: concerned customer

"Your application is very slow to use. I tried it from both my phone and my computer, and it's the same everywhere."

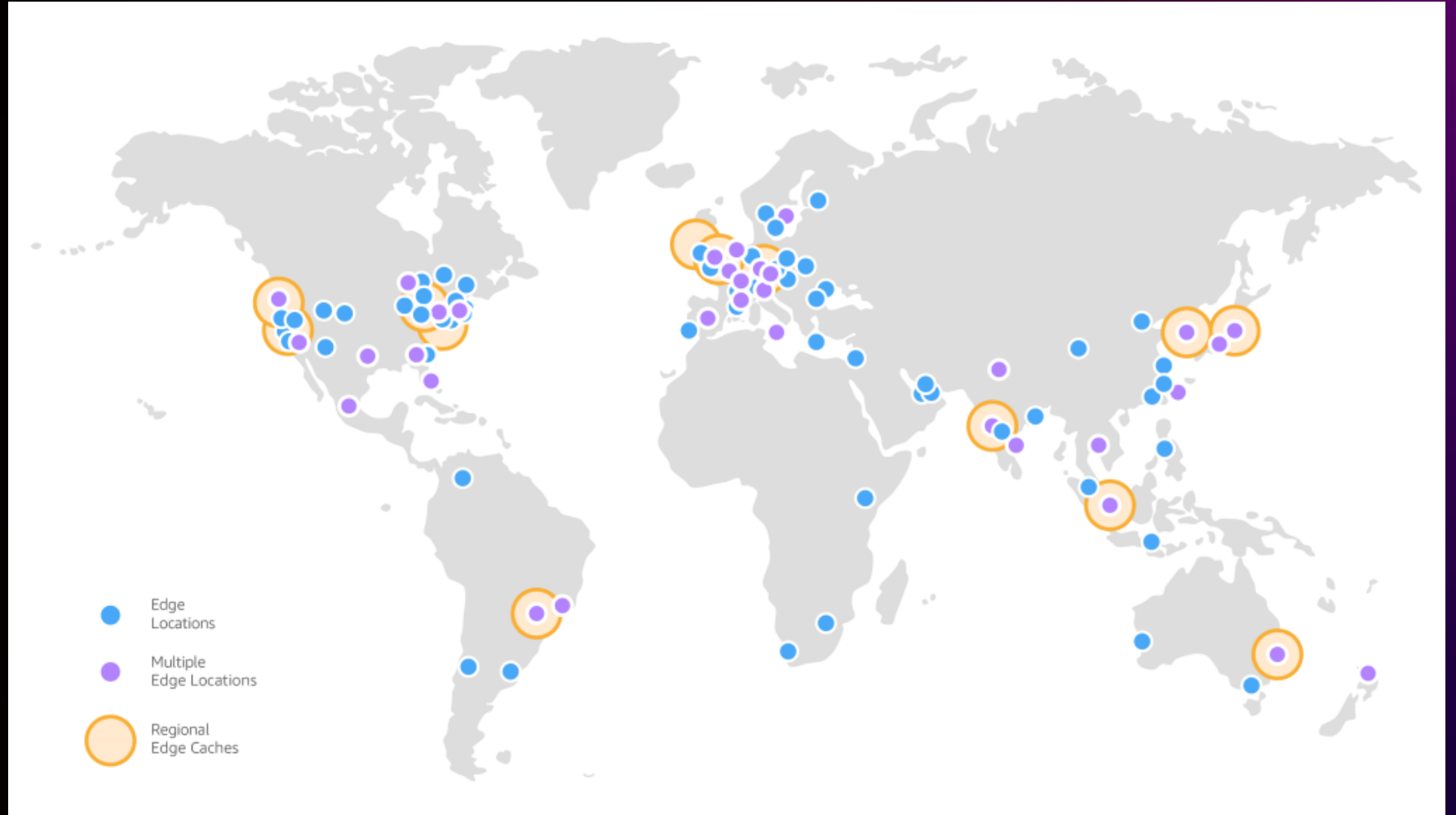
The dashboard is all green



CloudWatch graphs for the internet?



CloudWatch graphs for the internet?



It wasn't Sam's fault



After replying to several tickets, Sam discovers that all customers are using the same ISP

When checking the status page of that ISP, Sam finds a notice that part of their network is experiencing packet loss because of a fiber cut somewhere

Introducing CloudWatch Internet Monitor



Sam heard about CloudWatch Internet Monitor and decides to try it

Setting up an Internet Monitor is easy

- Quick setup process (just 3 steps)
- Observe traffic data within minutes (not hours)
- Receive alerts on the go with Amazon EventBridge integration

Sam gets an internet outage alert



aws Services Search for services, service features, Marketplace products, docs, and more

N. Virginia MyRole/AWSUser @ 0123-4567-8901

Events map last updated 3 minutes ago.

Reset Zoom

Las Vegas, NV, USA
Availability score: 94.50%
Performance score: ⚠ 60.00%
End-user traffic impacted: 1.78% impact to overall traffic, 40% impact at current client location

Health events (4) [Info](#) Actions ▼

Lists health events for your application. For example, traffic impact of N% for a health event for Location C and ISP A means that N% of traffic from Service location towards Location C over ISP A is experiencing an availability or performance drop. (Note that when service location is a Region, measurements and events represent connectivity at a Regional level, not between end-user locations and Availability Zones.)

Filter by Client location, Client ISP, Analysis, or AWS source

	Client location ▼	Client network ▼	Service location ▼	Traffic impact ▲	Impact type ▼
☐	Las Vegas, NV, USA	Comcast (AS 7922)	us-east-1	1.78%	Internet performance issue
☐	Seattle, WA, USA	ISP B network name (AS 45678)	us-west-1	1.07%	Internet performance issue
☐	Tijuana, Mexico	ISP C network name (AS 7890)	ap-southeast-1	0.97%	AWS availability issue
☐	Bangkok, Thailand	ISP D network name (AS 54321)	Edge	0.54%	Internet performance issue

Feedback

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CloudWatch Internet Monitor

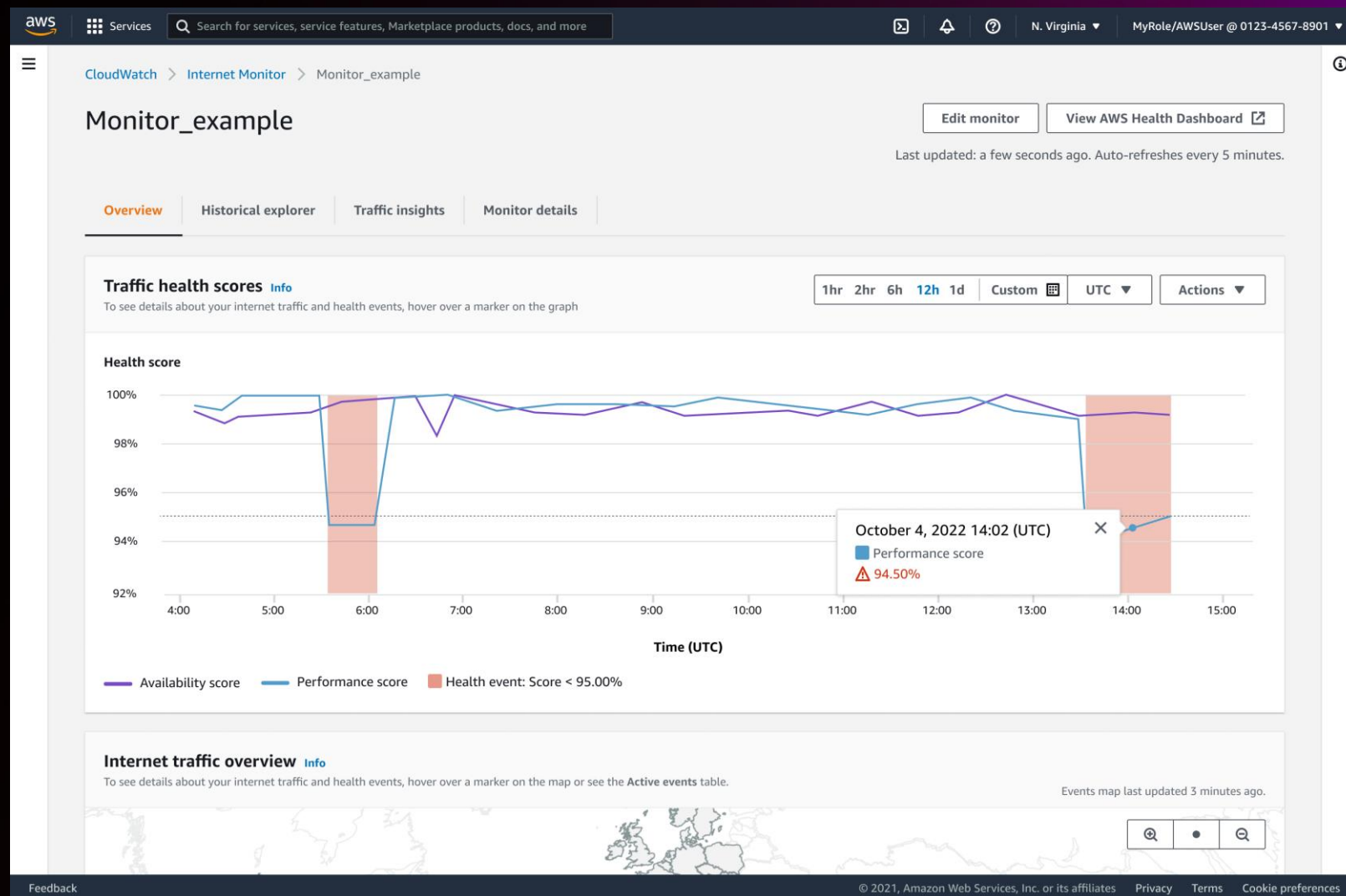


A customer calls

Sam's deep dive into CloudWatch Internet Monitor



Overview: Performance and availability score



Live tab: Availability and performance score



aws

Services

Search for services, service features, Marketplace products, docs, and more

MyRole/AWSUser @ 0123-4567-8901

N. Virginia

Events map last updated 3 minutes ago.

Las Vegas, NV, USA

Availability score
94.50%

Performance score
 60.00%

End-user traffic impacted
1.78% impact to overall traffic
40% impact at current client location

Location impacted by an active event

☐ Client locations with internet traffic

Health events (4)

Info

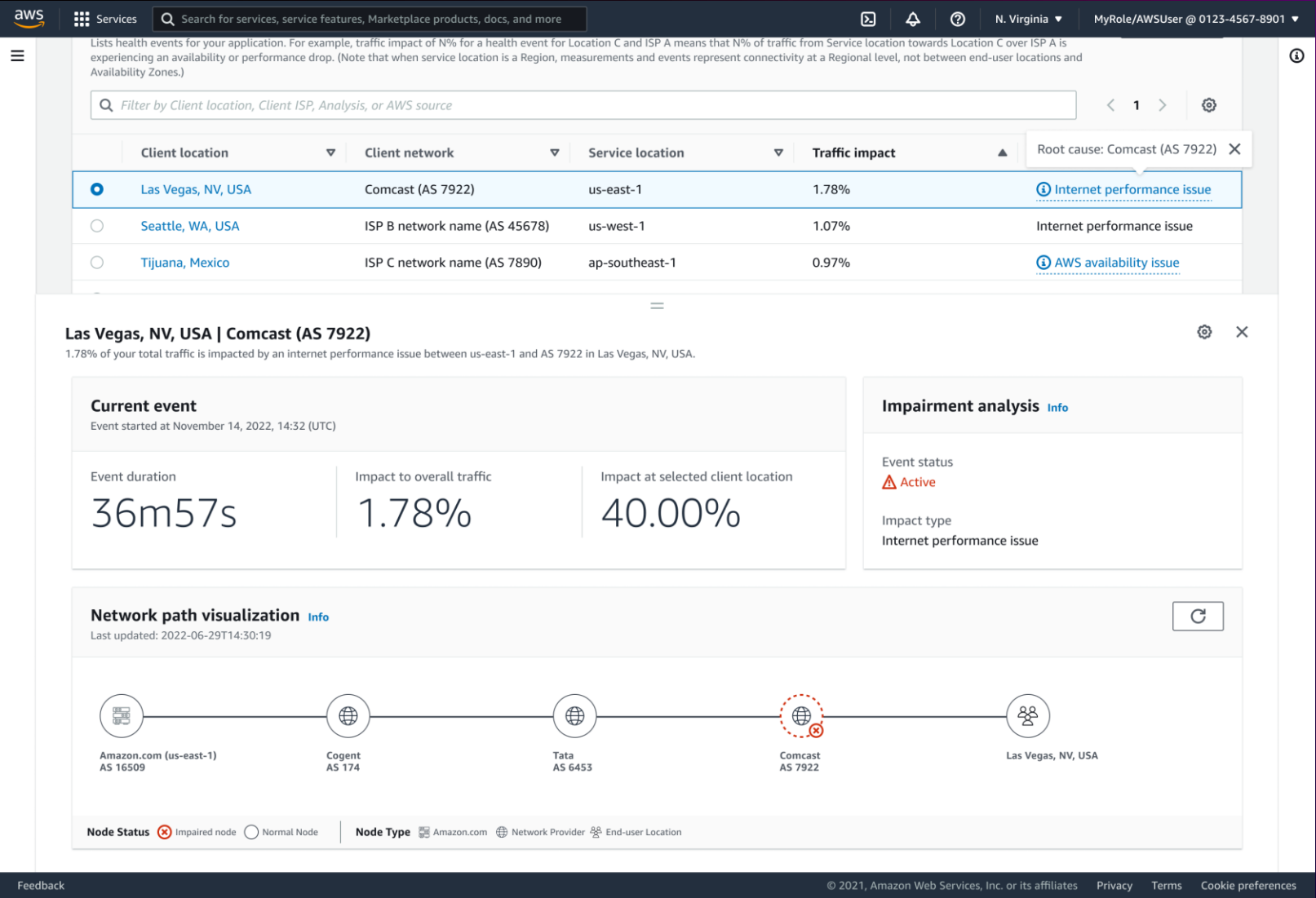
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Overview: Network path visualization



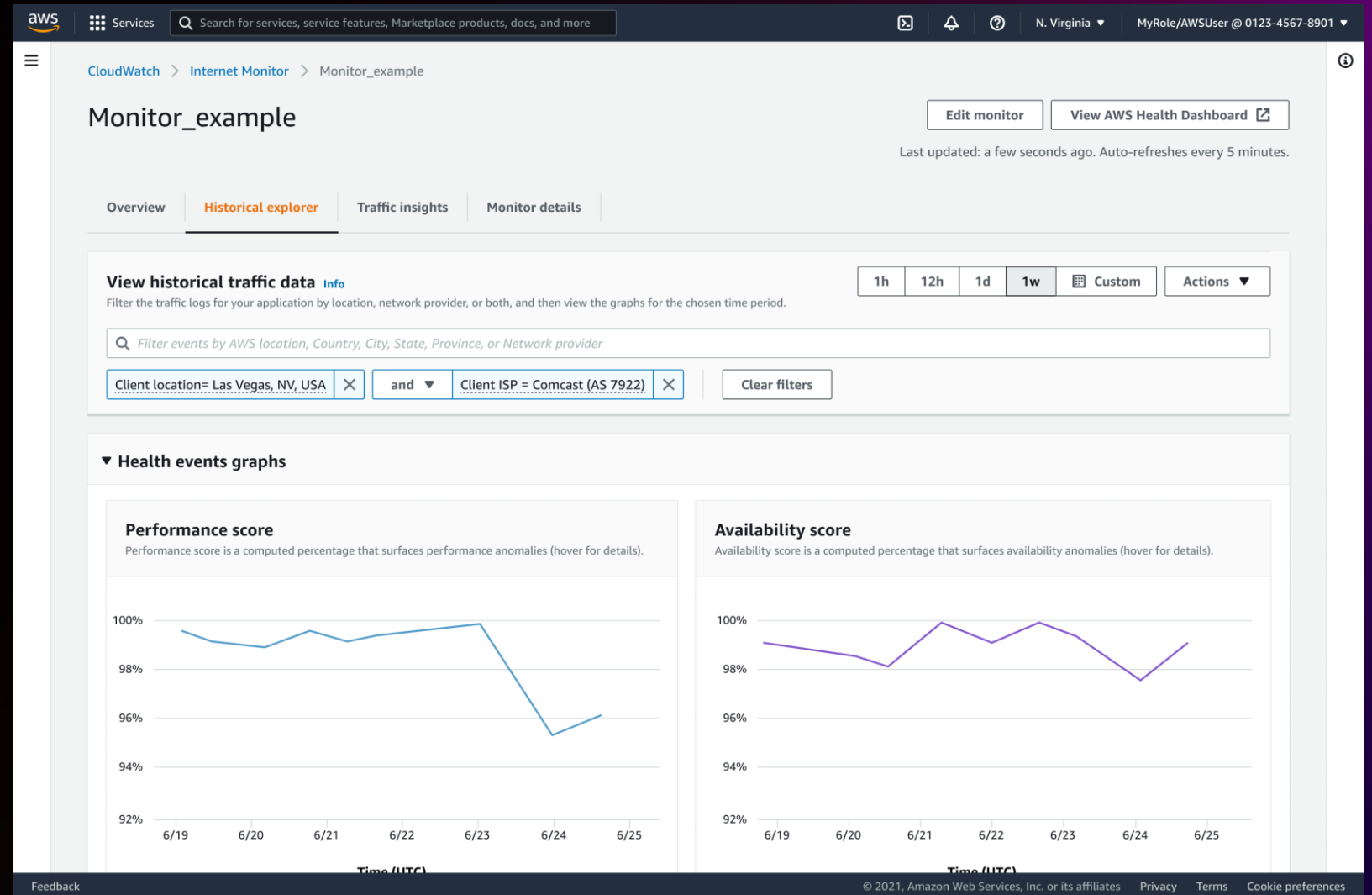


Issue identified

With the internet monitor dashboard, Sam identified the issue and its impact on end user traffic for their customers; he exports the data and proceeds to engage the ISP to work on addressing the problem

Meanwhile, Sam decides to explore historical data about the issue

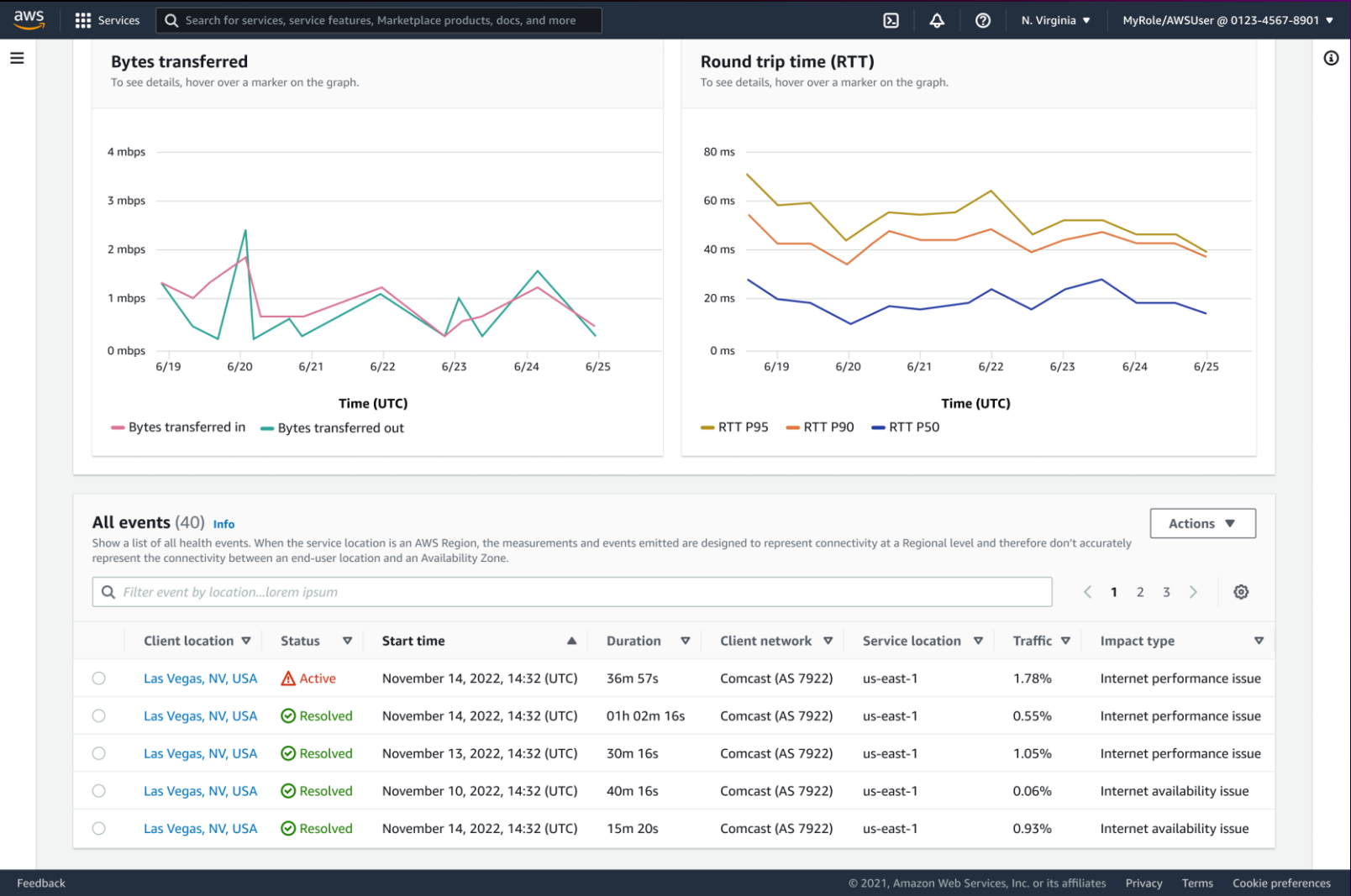
Historical explorer



Historical explorer tab: Performance score, Availability score



Historical explorer: All events table

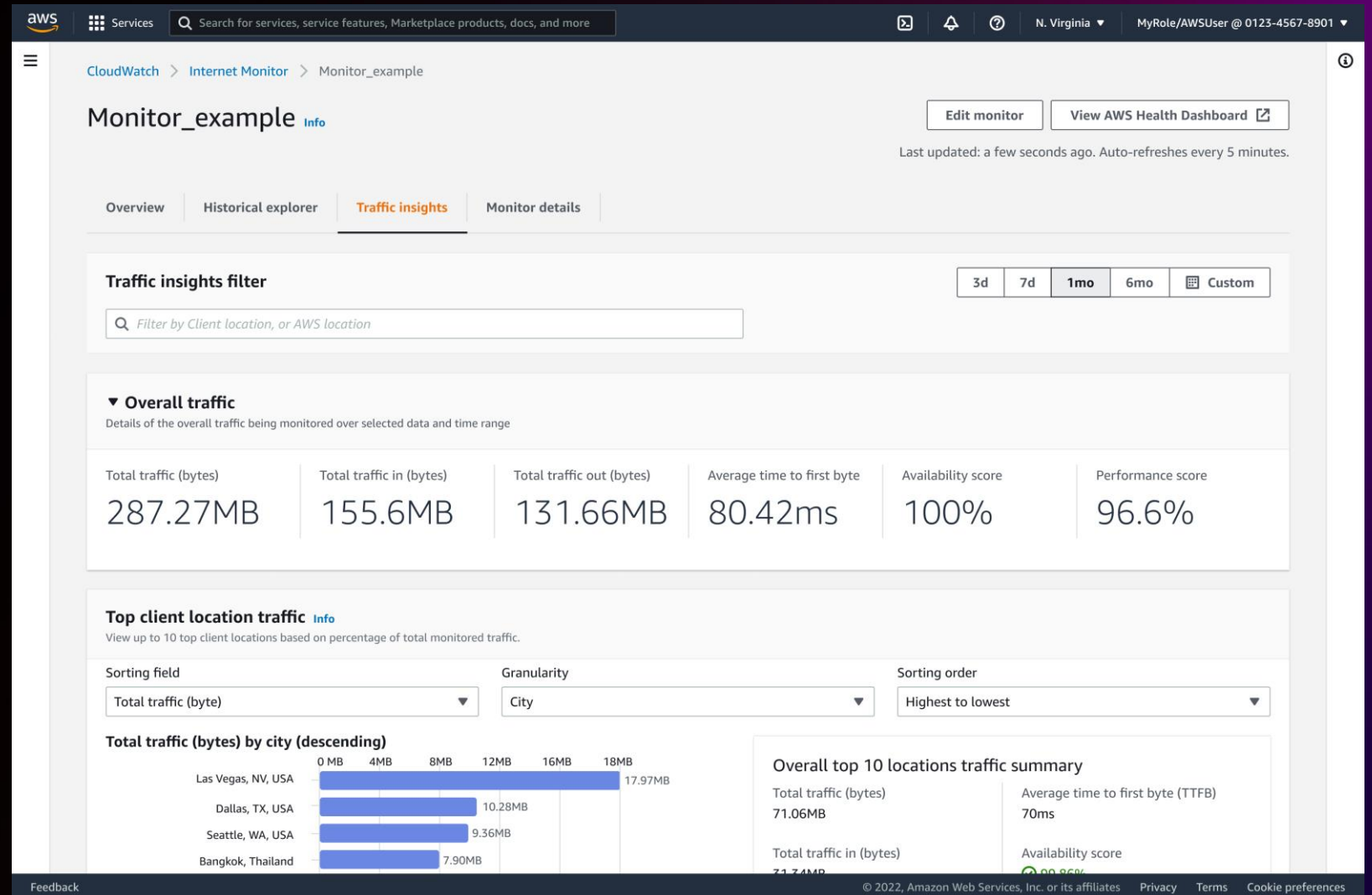




Trends understood

With the internet monitor historical explorer tab, Sam filtered and searched through previous log data to further understand the context and patterns of an issue

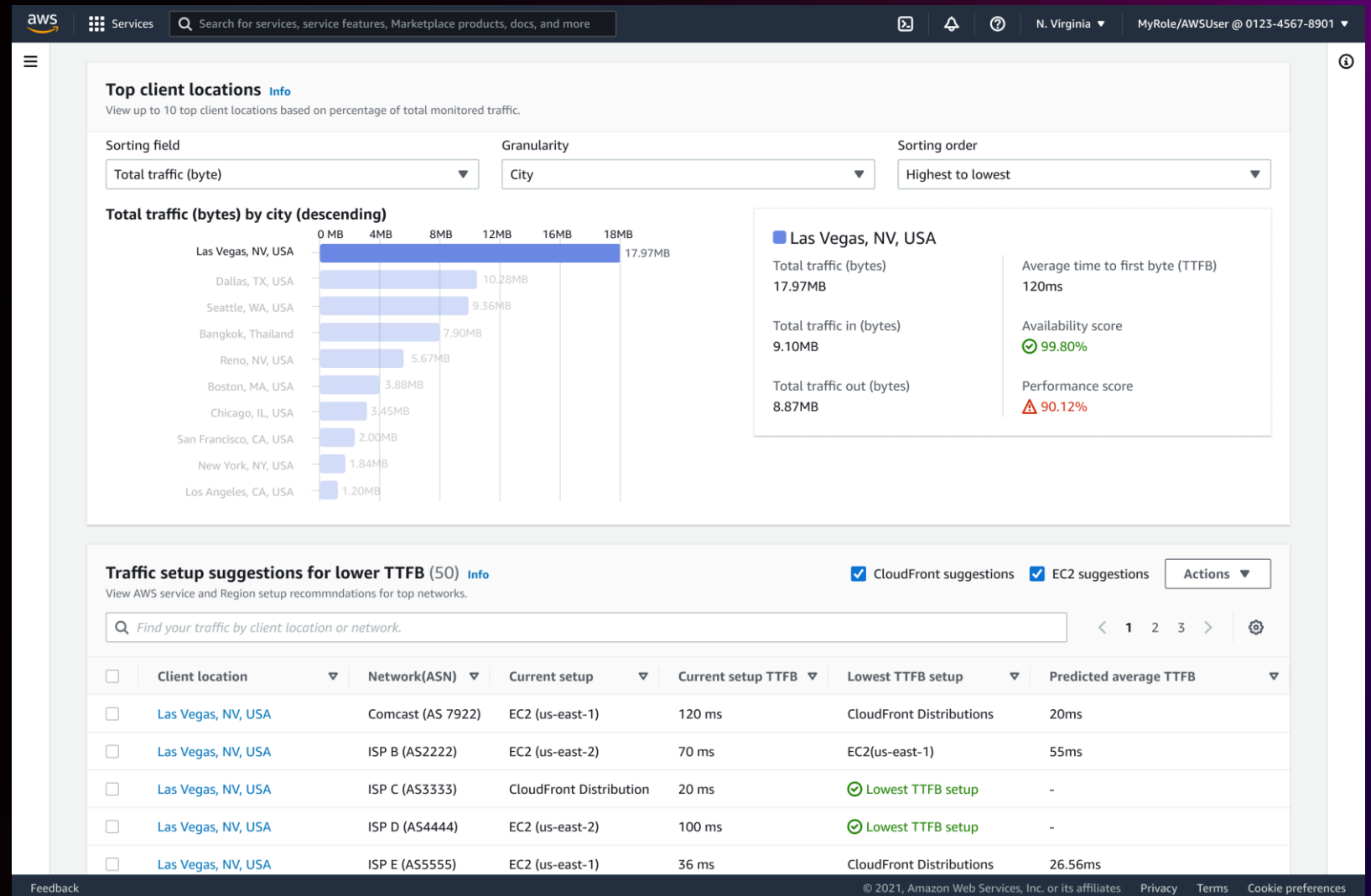
Traffic insights



Traffic insights tab: Overall insights



Traffic insights: Top client locations



Traffic insights tab: Overall insights

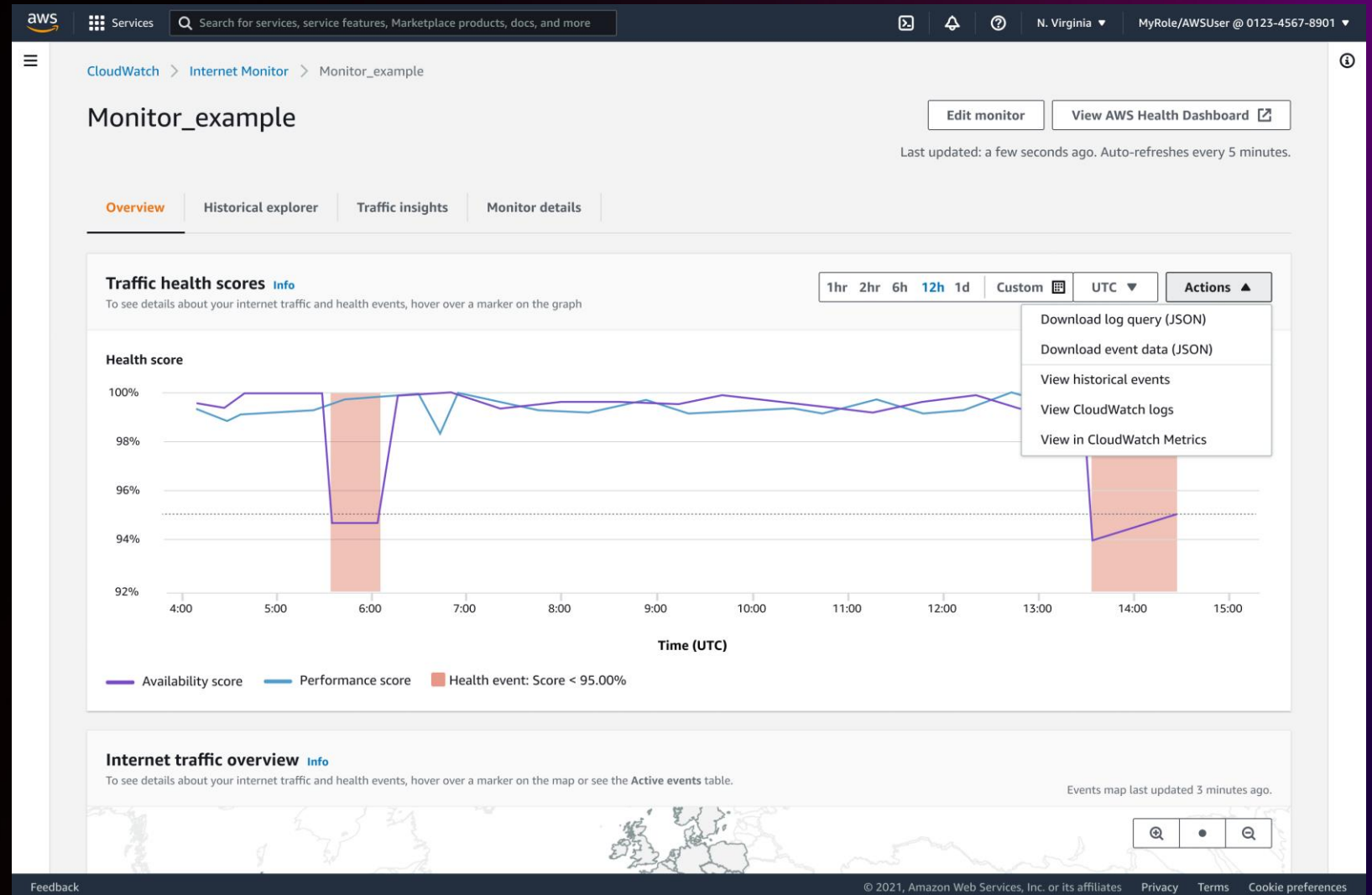




Architectural recommendations received

By exploring traffic insights, architectural recommendations, and the time to first byte predictions, Sam feels confident about proposing an architectural change as a long-term solution to help prevent this issue in the future

CloudWatch logs and metrics



Traffic insights tab: Overall insights



EventBridge integration

The screenshot shows the AWS CloudWatch console for an Internet Monitor named 'Monitor_example'. The 'Monitor details' tab is active, displaying the monitor's name, resource ID, ARN, and status (Active). Below this, the 'Monitored resources (26)' section shows a table of resources being monitored.

Monitor details

Monitor name Monitor_example	Monitor Amazon resource name (ARN) arn:aws:phoebe:us-west-2:123456789012:monitor/Monitor_example	Monitored resource type VPC, CloudFront Distributions	Status Active
Resource ID 1234567890abcdef0		Total resources 26	EventBridge alerts Go to EventBridge

Monitored resources (26)

Filter by resource name, resource ID, AWS Location, or resource type

Resource name	Resource ID	Service location	Resource type	Account ID
CloudFront resource name	resource###	CloudFront-Edge	CloudFront	000000000000
CloudFront resource name	resource###	CloudFront-Edge	CloudFront	000000000000
Resource name	vpc-#####1	us-west-2	VPC	000000000000
Resource name	vpc-#####2	us-east-1	VPC	000000000000
Resource name	vpc-#####3	us-east-1	VPC	000000000000
Resource name	vpc-#####4	us-west-2	VPC	000000000000
Resource name	vpc-#####5	us-west-2	VPC	000000000000
Resource name	vpc-#####6	us-west-2	VPC	000000000000

Traffic insights tab: Overall insights



Architecture

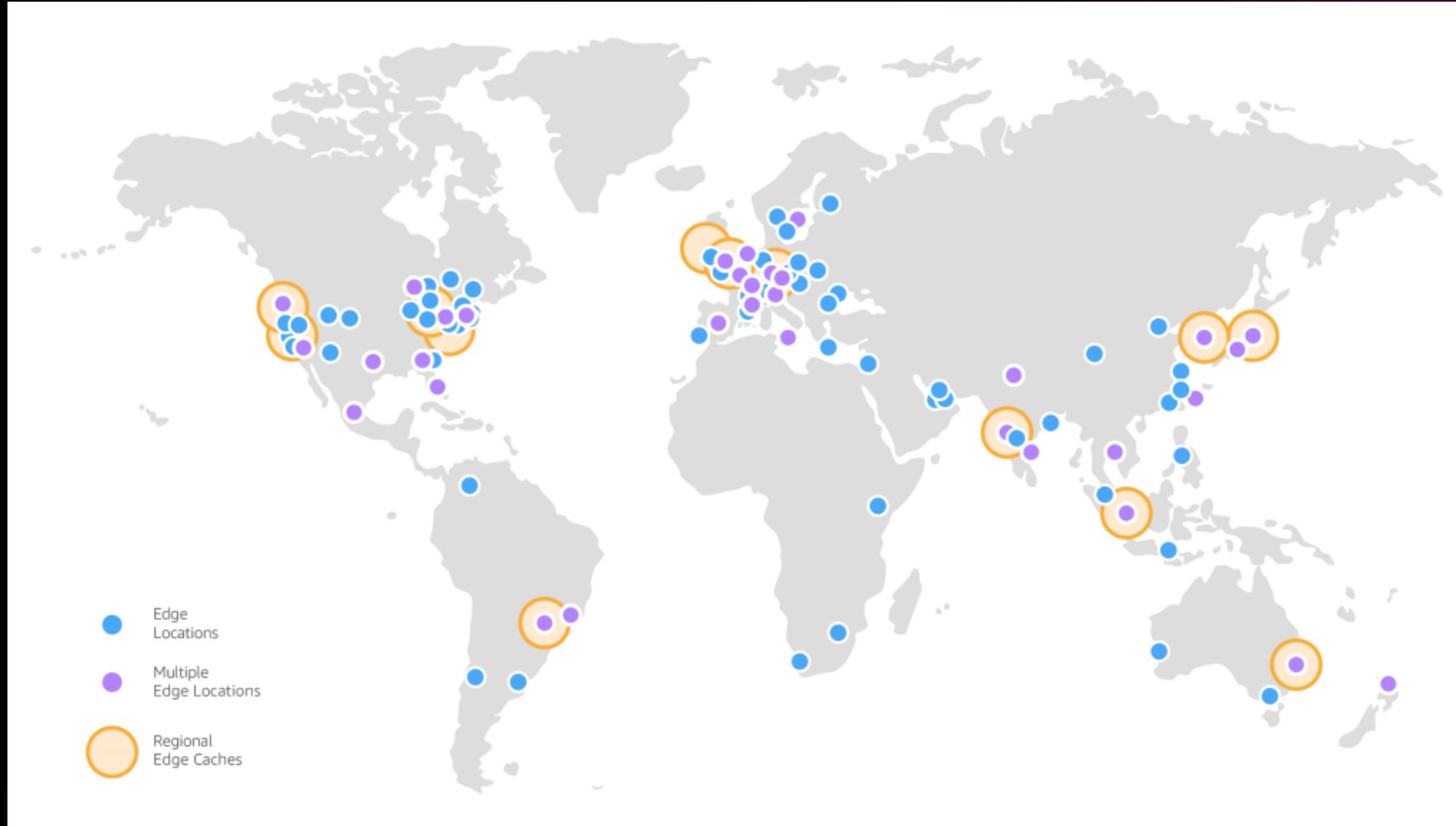


CloudWatch Internet Monitor architecture

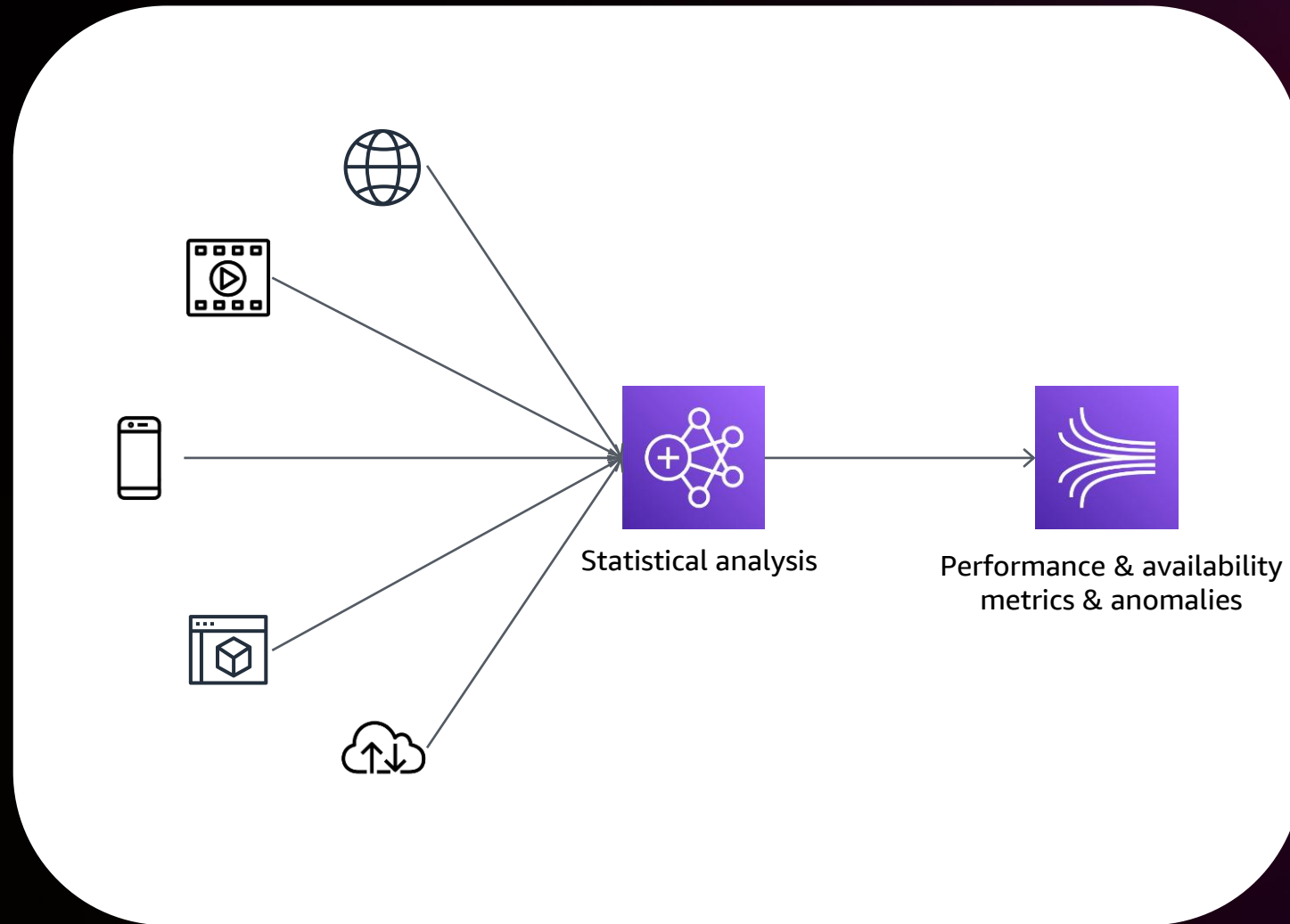
It's all about the data



CloudWatch Internet Monitor architecture



CloudWatch Internet Monitor architecture

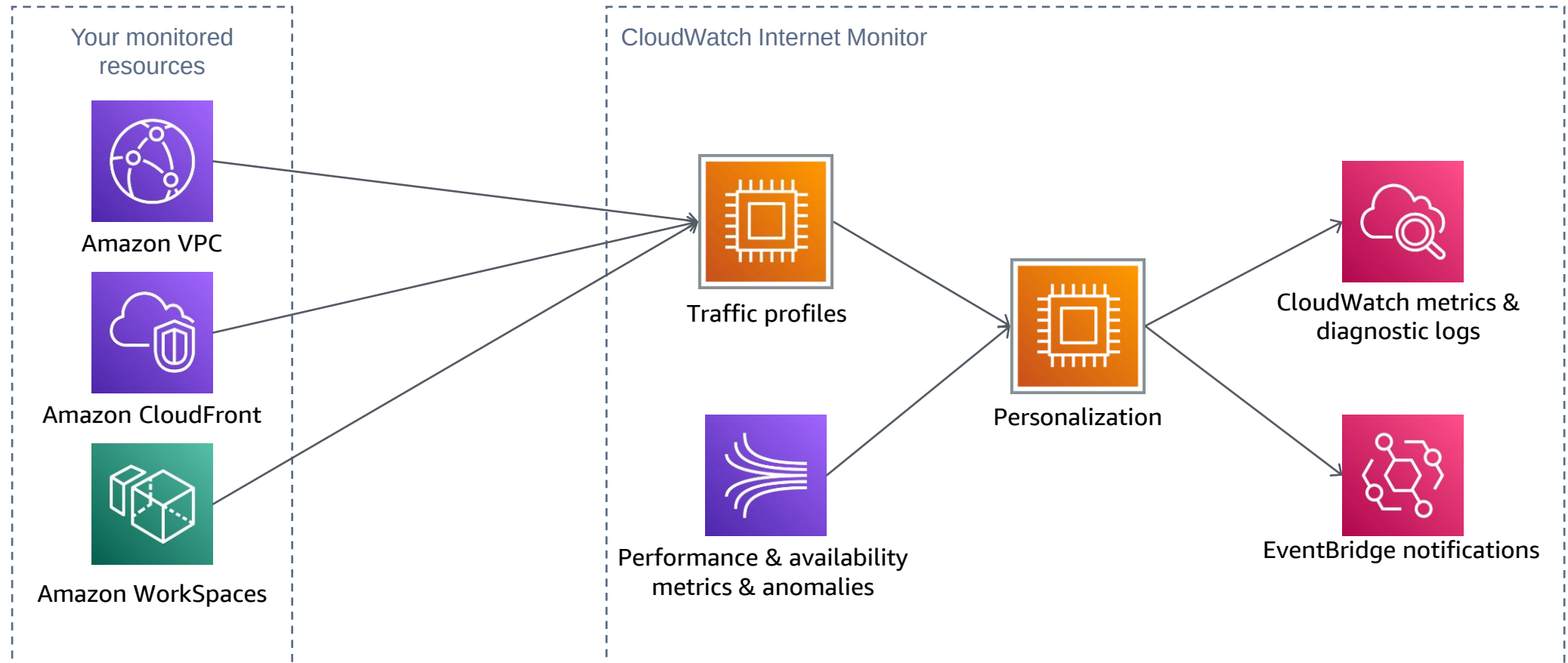


CloudWatch Internet Monitor architecture

And then it's about personalization



CloudWatch Internet Monitor architecture



Interested in learning more?

Amazon CloudWatch Internet Monitor blogs

<https://aws.amazon.com/blogs/aws/cloudwatch-internet-monitor-end-to-end-visibility-into-internet-performance-for-your-applications/>

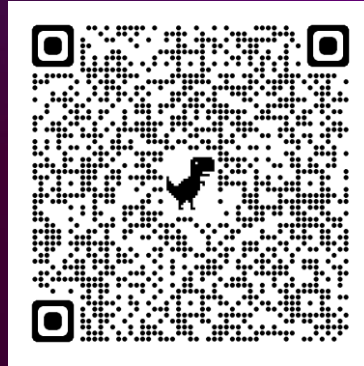
Amazon CloudWatch Internet Monitor What's New Post

<https://aws.amazon.com/about-aws/whats-new/2022/11/amazon-cloudwatch-internet-monitor-preview/>

Amazon CloudWatch Internet Monitor documentation

<https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/CloudWatch-InternetMonitor.html>

Using Internet Monitor in the Amazon CloudWatch User Guide



Thank you!



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